

# 2

## AVERAGES & AGES

### 2.1. What is Average?

The concept of average is equal distribution of the overall value among all the things or persons present there. So the formula for finding the average is as follows:

$$\text{Average, } A = \frac{\text{Total of all things, } T}{\text{Number of things, } N}$$

$$\text{Therefore, Total, } T = AN$$

If any person joins a group with more value than the average of the group then the overall average increases. This is because the value in excess than the average will also be distributed equally among all the members.

Similarly when any value less than the average joins the group the overall group decreases as the deficit is divided equally among all the people present there.

#### Example:

Consider three people A, B and C with total of Rs. 30/-. Their average becomes Rs. 10/- for each. If another person D joins them with Rs. 50/- then he has Rs. 40/- more than actual average of Rs. 10/-.

So this Rs. 40/- will get distributed among those four and each gets Rs. 10/-. Thus the average becomes Rs. 20/- each.

#### Example:

The average age of a class of 30 students is 12. If the teacher is also included the average becomes 13 years. Find the teacher's age.

#### Solution:

- When the teacher is included there are totally 31 members in the class and the average is increased by 1 year. This means that everyone got 1 extra year after distributing the extra years of the teacher.
- So extra years of the teacher are as follow:  $31 \times 1 = 31$  years.
- Age of the teacher = actual avg + extra years =  $12 + 31 = 43$  years.

